

- a. **Emily** 要么是外科医生，要么是律师

$\text{Occupation}(\text{Emily}, \text{Surgeon}) \vee \text{Occupation}(\text{Emily}, \text{Lawyer}) \wedge \neg(\text{Occupation}(\text{Emily}, \text{Surgeon}) \wedge \text{Occupation}(\text{Emily}, \text{Lawyer}))$

- b. **Joe** 是个演员，但他还有另外的工作

$\text{Occupation}(\text{Joe}, \text{Actor}) \wedge \exists o (o \neq \text{Actor} \wedge \text{Occupation}(\text{Joe}, o))$

- c. 所有外科医生都是医生

$\forall p (\text{Occupation}(p, \text{Surgeon}) \rightarrow \text{Occupation}(p, \text{Doctor}))$

- d. **Joe** 没有律师（即，他不是任何律师的客户）

$\forall p (\text{Occupation}(p, \text{Lawyer}) \rightarrow \neg \text{Customer}(\text{Joe}, p))$

- e. **Emily** 的老板是个律师

$\exists p (\text{Boss}(p, \text{Emily}) \wedge \text{Occupation}(p, \text{Lawyer}))$

- f. 有个律师的客户全都是医生

$\exists p (\text{Occupation}(p, \text{Lawyer}) \wedge \forall q (\text{Customer}(q, p) \rightarrow \text{Occupation}(q, \text{Doctor})))$

- g. 每个外科医生都有律师

$\forall p (\text{Occupation}(p, \text{Surgeon}) \rightarrow \exists q (\text{Occupation}(q, \text{Lawyer}) \wedge \text{Customer}(p, q)))$